

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE CODE: CSA51
Network Security

COURSE NAME: Cryptography and

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5
25

TOTAL MARKS:

Annexure A
PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

I Y.Naga Tharuni/192372338 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Naga Tharuni*

Annexure A
PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Design pseudocode for a stateful firewall system that tracks active connections and filters packets based on session state. Evaluate how state tracking improves security compared to stateless filtering.

Stateful Firewall System

Pseudocode

ALGORITHM StatefulFirewall

INPUT: Incoming or outgoing network packet

OUTPUT: ACCEPT or DROP

BEGIN

 Create empty ConnectionTable

 FOR each packet received DO

 Extract:

 SourceIP

 DestinationIP

 SourcePort

 DestinationPort

 Protocol

 TCP_Flags

 ConnectionKey ←

 (SourceIP, DestinationIP, SourcePort, DestinationPort, Protocol)

 IF ConnectionKey exists in ConnectionTable THEN

 State ← ConnectionTable[ConnectionKey]

 IF State is ESTABLISHED OR State is RELATED THEN

 Update connection timeout

 Update connection state if required

 ACCEPT packet

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ELSE
    DROP packet
ENDIF

ELSE

    IF packet starts a valid new connection AND
        packet satisfies firewall security rules THEN

        Add ConnectionKey to ConnectionTable
        Set State ← NEW
        ACCEPT packet

    ELSE
        DROP packet
    ENDIF

ENDIF

Remove expired connections from ConnectionTable

ENDFOR

END

```

Evaluation

A stateful firewall maintains information about active network connections. It can determine whether a packet belongs to an established, legitimate session.

Compared with a stateless firewall, which examines each packet independently, state tracking provides better security because:

- It can block unexpected packets that do not belong to an existing session.
- It helps prevent unauthorized connection attempts.
- It can identify suspicious packets that violate the expected session state.
- It reduces the need to evaluate every packet only using static rules.

Therefore, stateful filtering provides better context-aware protection than stateless filtering, although it requires additional memory and processing for maintaining the connection table.

2. Design pseudocode for a signature-based malware detection system that scans files and compares them against a known signature database. Evaluate its effectiveness in detecting known threats and its limitations against new attacks.

Signature-Based Malware Detection System

Pseudocode

ALGORITHM SignatureBasedMalwareDetection

INPUT: File

OUTPUT: SAFE or MALWARE DETECTED

BEGIN

 Load KnownSignatureDatabase

 Calculate FileHash $\leftarrow$ Hash(File)

 IF FileHash matches a signature in KnownSignatureDatabase THEN

 Display "Malware Detected"

 Identify malware name and type

 Quarantine File

 RETURN MALWARE DETECTED

 ENDIF

 Extract file patterns and byte sequences

 FOR each Signature in KnownSignatureDatabase DO

 IF FilePattern matches Signature THEN

 Display "Known Malware Detected"

 Quarantine File

 RETURN MALWARE DETECTED

 ENDIF

 ENDFOR

 Display "No Known Malware Detected"

 RETURN SAFE

END

Evaluation

Signature-based detection is highly effective for known malware because previously identified malicious files can be recognized quickly using hashes or known patterns.

Advantages:

- High accuracy for known threats.
- Fast scanning.
- Usually produces fewer false positives when signatures are reliable.
- Easy to update by adding new malware signatures.

Limitations:

- Cannot reliably detect completely new malware without a known signature.
- Polymorphic malware can change its appearance to avoid signature matching.
- Zero-day attacks may remain undetected.
- The signature database must be regularly updated.

Thus, signature-based detection is effective for known threats, but it should be combined with behavioral or machine-learning detection for better protection against new attacks.

3. Design pseudocode for a distributed denial-of-service (DDoS) detection **system** using traffic threshold analysis. Evaluate how the algorithm differentiates between legitimate high traffic and attack traffic.

DDoS Detection System Using Traffic Threshold Analysis

Pseudocode

ALGORITHM DDoSDetection

INPUT: Network traffic statistics

OUTPUT: NORMAL TRAFFIC or POSSIBLE DDoS ATTACK

BEGIN

 Set TrafficThreshold

Set PacketRateThreshold
Set SourceThreshold

Continuously monitor network traffic

Every monitoring interval DO

TotalPackets $\leftarrow$ Count incoming packets
PacketRate $\leftarrow$ TotalPackets / MonitoringInterval
UniqueSources $\leftarrow$ Count unique source IP addresses
BandwidthUsage $\leftarrow$ Calculate bandwidth usage

IF PacketRate > PacketRateThreshold OR
BandwidthUsage > TrafficThreshold THEN

Analyze traffic characteristics

IF UniqueSources is extremely high AND
traffic contains abnormal patterns THEN

Generate DDoS Alert
Apply rate limiting
Block or filter suspicious traffic
Log attack information

ELSE
Mark traffic as POSSIBLE LEGITIMATE SPIKE
Continue monitoring
ENDIF

ELSE

Mark traffic as NORMAL

ENDIF

END EVERY

END

Evaluation

The algorithm identifies possible DDoS attacks by comparing traffic measurements with predefined thresholds.

To differentiate legitimate high traffic from an attack, it can consider:

- Packet rate.
- Bandwidth consumption.
- Number of source IP addresses.
- Suddenness and duration of the traffic increase.
- Abnormal packet patterns.

A legitimate event, such as a product launch, may produce high traffic but usually has recognizable and valid application behavior. A DDoS attack may show an unusually rapid increase in traffic, many suspicious sources, repeated requests, or abnormal packet patterns.

However, fixed thresholds can produce false positives during genuine traffic spikes. Adaptive thresholds and additional behavioral analysis can improve accuracy.

4. Design pseudocode for a network intrusion detection system using machine learning classification. Evaluate how feature selection and training data impact detection accuracy.

Network Intrusion Detection Using Machine Learning Classification

Pseudocode

ALGORITHM MLNetworkIntrusionDetection

INPUT: Network traffic data

OUTPUT: NORMAL or INTRUSION

BEGIN

 Collect labeled network traffic dataset

 Preprocess dataset:

 Remove invalid records

 Handle missing values

 Normalize numerical features

Select important features such as:

- PacketRate
- SourceIP
- DestinationIP
- SourcePort
- DestinationPort
- Protocol
- PacketSize
- ConnectionDuration
- NumberOfConnections

Split dataset into:

- TrainingData
- TestingData

Train MachineLearningModel using TrainingData

Evaluate model using TestingData

Calculate:

- Accuracy
- Precision
- Recall
- FalsePositiveRate

IF performance is acceptable THEN

- Deploy trained model

ELSE

- Improve feature selection or training data
- Retrain model

ENDIF

FOR each new network connection DO

- Extract required features

- Prediction $\leftarrow$ MachineLearningModel(features)

- IF Prediction = INTRUSION THEN

- Generate intrusion alert
 - Log connection details

- ELSE

- Allow normal monitoring

ENDIF

ENDFOR

END

Evaluation

Feature selection has a major impact on detection accuracy. Relevant features help the model distinguish normal traffic from malicious traffic. Irrelevant or redundant features can increase complexity and reduce performance.

Training data is equally important. A dataset containing diverse and correctly labeled examples helps the model learn different attack patterns.

Poor training data can cause:

- High false-positive rates.
- High false-negative rates.
- Poor detection of attacks not represented in the dataset.
- Overfitting to specific traffic patterns.

Therefore, a good IDS should use relevant features, diverse training data, proper preprocessing, and regular model evaluation.

5. Design pseudocode for a secure network access control system that authenticates users and enforces role-based access policies. Evaluate how access control mechanisms enhance overall network security.

Secure Network Access Control System

Pseudocode

ALGORITHM SecureNetworkAccessControl

INPUT: Username, Password, RequestedResource, Action

OUTPUT: ACCESS GRANTED or ACCESS DENIED

BEGIN

 Receive Username and Password

Retrieve user account from UserDatabase

IF user account does not exist THEN

Deny access

Log failed authentication

RETURN ACCESS DENIED

ENDIF

Verify Password

IF Password is invalid THEN

Increment failed login counter

Log failed authentication

IF failed login counter exceeds limit THEN

Temporarily lock account

ENDIF

RETURN ACCESS DENIED

ENDIF

Perform additional authentication if required

Set UserRole $\leftarrow$ authenticated user's role

Retrieve permissions associated with UserRole

IF RequestedResource is allowed for UserRole AND

Action is allowed by the assigned permissions THEN

Grant access

Log successful access

RETURN ACCESS GRANTED

ELSE

Deny access

Log unauthorized access attempt

RETURN ACCESS DENIED

ENDIF

END

Evaluation

Access control improves network security by ensuring that users can access only the resources and operations permitted for their roles.

Role-Based Access Control (RBAC) provides:

- Authentication of users.
- Authorization based on user roles.
- Least-privilege access.
- Protection of sensitive resources.
- Reduced risk of unauthorized modifications.
- Logging and monitoring of access attempts.

For example, an administrator may be allowed to modify network configurations, while a normal employee may only access required applications.

Therefore, combining strong authentication with role-based authorization provides an important security layer and reduces the possibility of unauthorized access to network resources.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient